Response to Reviewer Comments

Response to Reviewer #1

We sincerely thank the reviewer for the insightful comments and constructive suggestions. We have carefully addressed each point and revised the manuscript accordingly. Below we provide detailed point-by-point responses.

Major Comments

1. **The causal interpretation in the mediation analysis relies on strong assumptions and may be under-identified given potential confounders and timing frictions; clarify identification assumptions and timing choices, and provide a brief narrative explaining why the observed patterns are consistent with the intended causal pathway.**

   Response: We acknowledge the reviewer's important concern about causal identification in our mediation analysis. We have revised the manuscript to explicitly address these issues:

   1. Identification Assumptions: We have clearly stated that our mediation analysis relies on three key assumptions:

   - No unmeasured confounding of the GPR → Ambiguity relationship
   - No unmeasured confounding of the Ambiguity → Returns relationship (conditional on controls)
   - Correct temporal ordering (GPR affects ambiguity before ambiguity affects returns)

   2. Timing Clarification: We have specified that:

   - GPR index is constructed from news articles published at 00:00 UTC (08:00 Beijing time)
   - Ambiguity is calculated from intraday returns during Chinese trading hours (09:30-15:00 Beijing time)

- This timing ensures GPR news can affect investor beliefs before trading begins

3. Narrative Explanation:
We have added the following narrative to strengthen causal interpretation:

"The observed mediation pattern aligns with our causal story through several mechanisms. First, geopolitical news creates model uncertainty as investors reassess their beliefs about return distributions. This is reflected in increased cross-entropy divergence. Second, this heightened ambiguity leads to risk premiums as ambiguity-averse investors demand compensation for model uncertainty, resulting in lower contemporaneous returns. The temporal sequence—morning GPR news affecting midday ambiguity calculations, which then influence end-of-day returns—supports this causal pathway."

Beyond the mediation framework, we have reinforced identification with complementary checks: an instrumental-variable design using Non-Asian GPR shocks, placebo tests using non-geopolitical macro headlines, and sensitivity analyses across control sets (market, VIX, rates). Bootstrap confidence intervals and Newey–West inference further ensure robust estimation in the presence of short-run serial correlation. These additions clarify the assumptions, timing, and empirical evidence consistent with the intended causal pathway.

2. **The KL-based ambiguity measure depends on the choice of benchmark distributions and implicit stationarity across regimes; explain the rationale for benchmark selection and stationarity, and add a short methodological note discussing how plausible alternative choices would affect interpretation.**

Response: We thank the reviewer for highlighting this important methodological point. We have made the following additions:

1. Benchmark Selection Rationale: We have clarified that:

- We use rolling windows of 252 trading days (approximately 1 year) for benchmark distributions
- This window balances stability and responsiveness
- Windows are anchored to the start of each quarter to reduce look-ahead bias
- We exclude extreme volatility periods (>3 standard deviations) from benchmarks

2. Stationarity Considerations:

We have acknowledged that stationarity is a strong assumption and added:
"While our approach assumes local stationarity within benchmark windows, we test this assumption by examining ambiguity persistence across structural breaks. Results show ambiguity reverts slowly (half-life of approximately 15 trading days), suggesting temporary regime shifts are captured by our adaptive framework."

3. Methodological Note:

We have appended the following discussion:
"Alternative benchmark specifications yield qualitatively similar results. Using (i) expanding windows from 2010, (ii) regime-switching benchmarks, or (iii) minimum KL divergence with alpha parameter -1 instead of -infinity, the correlation with our baseline ambiguity measure exceeds 0.85. However, alternative measures may be more appropriate in specific contexts: expanding windows better capture long-term structural changes but may over-smooth recent shocks; regime-switching models explicitly account for different market states but require stronger parametric assumptions."
In addition, we have discussed alternative divergence choices (e.g., Jensen–Shannon divergence) and generalized-mean aggregations over benchmark sets. These extensions produce materially similar inference in our data, which supports the robustness of using minimum KL divergence as a parsimonious and interpretable baseline.

3. **Frequency and timing alignment between daily GPR, ambiguity, realized volatility, and market/stock returns could introduce measurement mismatch; clearly state timestamp conventions (local vs UTC), intraday aggregation rules, and lead–lag considerations, and comment on why the chosen alignment is appropriate.**

Response: We have added a dedicated "Data Timing and Alignment" subsection with explicit details:

1. Timestamp Conventions:

- GPR Index: Published at 00:00 UTC, converted to Beijing time (UTC+8) for alignment
- Ambiguity: Calculated at market close (15:00 Beijing time) using full-day intraday data
- Realized Volatility: Computed from 5-minute returns between 09:30-15:00 Beijing time
- Market Returns: CSI 300 Index calculated at 15:00 Beijing time close

2. Aggregation Rules:

- High-frequency returns are aggregated using sum of log returns
- Cross-entropy is computed using price levels, not returns
- Weekend and holiday effects are handled by carrying forward previous day's values

3. Lead–Lag Justification: "*We use contemporaneous alignment (t, t) for our main analysis after confirming that (t-1, t) specifications yield similar results. This is appropriate because: (1) GPR news typically arrives before market open; (2) ambiguity reflects updated beliefs formed during trading; (3) returns incorporate all information by day's end. Robustness tests with alternative alignments show our main results are not driven by timing choices.*"

We have complemented this with Granger-causality tests and residual autocorrelation diagnostics, confirming the temporal ordering and mitigating concerns about feedback from returns to same-day ambiguity. Event-study windows around major geopolitical episodes further illustrate immediate timing consistent with the chosen alignment.

4. **The SOE moderation may reflect sample composition or correlated institutional features rather than a distinct cushion from ambiguity; define SOE classification precisely, summarize sample characteristics, and discuss the mechanism in text with supportive descriptive evidence.**

Response: We appreciate the reviewer's suggestion to deepen the analysis of SOE heterogeneity. We agree that distinguishing between sample composition effects and institutional mechanisms is crucial for interpreting the moderation results. In the revised manuscript, we have formally defined our SOE classification based on CSRC standards and added a detailed discussion on the specific channels—government guarantees, policy support, and state financing—that provide the ambiguity cushion. We also provide a summary of sample characteristics below to demonstrate that the effect is not driven by size or industry concentration alone.

1. SOE Definition: We have added:
"*SOEs are classified based on the China Securities Regulatory Commission (CSRC) definition: firms where the state is the ultimate controlling shareholder (Over 50% direct voting shares (absolute control), or at least 30% ultimate cash-flow rights via a pyramidal ownership structure plus de facto control (e.g., majority board seats, voting agreements)). We verify this using the WIND database ownership classifications and cross-check with government stake announcements.* "

2. Sample Characteristics:
We provide the following summary of sample characteristics for the reviewer's reference. Please note that due to space constraints and word limits in the main text, we have opted to present this detailed breakdown here rather than including the full table in the article.

- SOEs: 34.2% of sample (923/2,700 firms)
- Average market cap: SOEs ¥124.3B vs Non-SOEs ¥45.8B
- Industry distribution: SOEs concentrated in energy (23%), finance (18%), materials (16%)
- Leverage: SOEs 1.82x vs Non-SOEs 1.34x debt-to-equity
- State shareholding: Average 58.3% for SOEs, 2.1% for non-SOEs

3. Mechanism Discussion: We have added descriptive evidence:
"The ambiguity cushion for SOEs likely operates through three channels: (1) Explicit government guarantees on debt obligations reduce default uncertainty; (2) Policy support during crises (e.g., targeted lending, tax relief) narrows outcome distributions; (3) Access to state-controlled financing creates a lower bound on firm value. Evidence from the 2020 COVID crisis confirms that SOEs benefited from stronger implicit guarantees, resulting in significantly more stable credit spreads and preferential access to liquidity support compared to non-SOEs (Allianz Trade, 2021)."

4. Alternative Explanations: We have tested and ruled out size effects by adding:
"Results hold after controlling for firm size and interaction terms. The SOE buffer persists within size quintiles, suggesting it's not merely a scale effect."

5. **The contribution would benefit from clearer positioning of the ambiguity channel relative to volatility-focused frameworks; refine the introduction and related work to emphasize incremental insight and practical relevance, and tighten the discussion of empirical results accordingly.**

Response: We sincerely thank the reviewer for this crucial observation. We agree that clearer positioning is essential to articulate the paper's theoretical and empirical value. We have significantly revised the Introduction and Literature Review to explicitly distinguish our ambiguity-focused framework from traditional volatility-based approaches. Specifically, we now frame the contribution around the distinction between Knightian uncertainty (ambiguity) and risk (volatility), demonstrating that ignoring ambiguity leads to a misattribution of pricing effects. We also tightened the results discussion to emphasize the economic magnitude of the ambiguity channel rather than just statistical significance.

1. Introduction Changes: To immediately clarify our contribution, we have inserted the following paragraph in the Introduction:

> *"While existing literature predominantly examines GPR through the lens of volatility (Bouri et al., 2021; Apergis et al., 2021), we demonstrate that ambiguity represents a distinct and economically significant transmission channel. Unlike volatility—which captures second-moment risk with known probabilities—ambiguity reflects Knightian uncertainty where probability distributions themselves are unknown. Our framework shows that: (i) GPR increases ambiguity as a separate channel alongside volatility; (ii) ambiguity explains 38.5% of GPR's effect on returns; and (iii) models ignoring ambiguity misattribute ambiguity's impact to volatility."*

2. Literature Review Enhancement:

We have added a new paragraph to the Literature Review to position our work against recent uncertainty studies:

"Our work extends the emerging literature on uncertainty transmission in several critical ways. First, while Baker et al. (2016) and Caldara and Iacoviello (2022) document that uncertainty matters for markets, they treat uncertainty as monolithic. By decomposing uncertainty into risk (volatility) and ambiguity, we reveal that ambiguity accounts for the majority of GPR's pricing effect. Second, unlike traditional proxy-based ambiguity measures (Brenner and Izhakian, 2018), our entropy-based approach captures real-time model uncertainty at daily frequency. Third, by examining the SOE heterogeneity, we uncover institutional mechanisms that mitigate ambiguity pricing—a dimension absent from volatility-focused studies."

3. Results Discussion Tightening:

In the results section, we have streamlined the statistical reporting to focus on incremental insights and economic significance:

- **Economic Significance:** We now explicitly state: "A one-standard-deviation increase in ambiguity reduces returns significantly, underscoring its economic relevance."
- **Comparative Magnitude:** We highlight that "including ambiguity alongside volatility improves model fit and reduces the volatility-only coefficient," underscoring why the ambiguity channel is non-negligible.

4. Practical Relevance:

To address the practical implications, we have added:

"Our findings have immediate practical implications. Portfolio managers using volatility-only hedging strategies remain exposed to ambiguity risk. During high-GPR periods, minimum-variance portfolios underperform relative to ambiguity-aware portfolios. Policy interventions focusing solely on volatility stabilization (e.g., circuit breakers) may miss the larger ambiguity transmission channel."

Minor Comments

1. **Clarify the sample period and exact trading days for all subsamples (pre-COVID vs COVID) with start/end dates.**

Response: We have added precise dates and expanded the description to ensure transparency and reproducibility:
"Full sample: January 3, 2018 to December 29, 2023 (1,458 trading days) Pre-COVID: January 3, 2018 to December 31, 2019 (485 trading days) COVID period: January 2, 2020 to December 29, 2023 (973 trading days)"

We have ensured consistency across tables, figures, and text. Trading-day counts are computed from the CSI 300 calendar excluding weekends and exchange-declared holidays; partial trading days and trading suspensions are excluded. Subsample boundaries align with major regime shifts (pre-pandemic vs pandemic), and we verified robustness to alternative breakpoints (e.g., early 2020 announcements). Data sources for calendar alignment and returns are WIND and official exchange release notes. This expanded description improves comparability across subsamples and clarifies how non-trading days are handled.

2. **Define all variables in table captions (e.g., "ΔAmbiguity", "RV") and specify units/frequencies.**

Response: We have updated all table captions with complete definitions and units/frequencies, and described transformation rules used in the analysis:

Example for Table 1: *"Variables: ΔGPR = daily change in Geopolitical Risk index; $\Delta Ambiguity$ = daily change in cross-entropy-based ambiguity measure; ΔRV = daily change in realized volatility (5-minute); Market Return = value-weighted A-share market return; VIX = CBOE Volatility Index; Term Spread = 10-year minus 1-year Chinese government bond yield; Risk-Free Rate = 3-month SHIBOR. All returns and spreads are in percentage points. Standard errors are Newey-West HAC robust with 5-day lag."*

We have standardized frequency annotations (daily vs monthly) and units (percentage points vs index units) across all tables. Where transformations are applied (e.g., log returns aggregated intraday, first differences for shocks), captions explicitly note the transformation to avoid ambiguity. Robustness panels that use alternative measures include an additional note clarifying the specific variant (e.g., alternative volatility proxy or entropy weighting).

3. **Increase figure/table font sizes and ensure captions state standard errors and lag choices (e.g., Newey–West 5-day) consistently.**

Response: We have implemented uniform readability and caption standards:

- Increased table font sizes from 9pt to 10pt and adjusted column spacing for improved readability at journal column widths
- Ensured all captions include: "Standard errors are Newey–West HAC robust with 5-day lag (unless otherwise specified)"
- Added: "*, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively"
- Verified axis label sizes and tick formatting for figures to maintain consistency with table readability standards

We have additionally verified legibility at typical journal column widths and adjusted spacing to improve readability without altering layout constraints.

4. **Ensure consistent notation for ambiguity and volatility betas across text, equations, and tables.**

Response: We have standardized notation and cross-checked usage throughout:

- Ambiguity beta: $\beta_{i,\text{AMB}}$ (not $\beta_{i,\mathcal{A}}$)
- Volatility beta: $\beta_{i,\text{VOL}}$ (not $\beta_{i,\text{RV}}$)
- Market beta: $\beta_{i,\text{MKT}}$ (consistent)
- Updated all tables and equations to use this notation and confirmed consistency in the mediation and robustness sections

We have cross-checked that these symbols are used uniformly in captions, footnotes, and the mediation section to prevent confusion.

5. **Add a brief algorithmic summary of the ambiguity measure in Appendix A with inputs/outputs.**

Response: We have added Algorithm 1 to Appendix A with inputs/outputs and procedural clarity:

Algorithm 1: Daily Ambiguity Calculation

Inputs: Intraday price vector $P_t^{(1:m)}$, historical return windows $\mathcal{W}_{t-252:t-1}$

Output: Ambiguity measure $\mathcal{A}_t$

Procedure:

1. Extract benchmark distributions: For each day $d \in \mathcal{W}_{t-252:t-1}$:

a. Compute intraday returns $r_d^{(i)} = \log(P_d^{(i)}/P_d^{(i-1)})$

b. Form empirical distribution π_d

2. Compute current distribution: μ_t from day t returns

3. Calculate KL divergences: $D_j = D_{KL}(\mu_t \| \pi_j)$ for all j

4. Return: $\mathcal{A}_t = \min_j D_j$

Complexity: $O(m \times k)$ where $m =$ intraday intervals, $k =$ benchmark days

We have ensured that this summary complements the mathematical definitions of Appendix A and cross-references the benchmark construction choices described in the main text, aiding replication and clarity.

6. **Report instrument strength and overidentification tests where IV is used.**

Response: We have added to the IV section and briefly summarized diagnostics: "*We report standard IV diagnostics indicating strong instruments and validity: first-stage statistics exceed conventional thresholds, and overidentification tests fail to reject instrument exogeneity.*"

We have also summarized weak-instrument robustness checks (e.g., conditional F statistics) and provided references to diagnostic tables in the appendix for transparency. This enrichment clarifies instrument relevance and exogeneity and aligns with best practice.

7. **Citations: include more up-to-date references and situate comparisons to contemporary uncertainty proxies (e.g., EPU and recent geopolitical uncertainty measures).**

Response: We have added recent references and integrated them into the comparative discussion:

- Bouri et al. (2024) on geopolitical risk, EPU, and volatility drivers across quantiles

- Liu et al. (2024) on geopolitical risk and green bond markets
- Zhang et al. (2023) on geopolitical risk and stock market volatility
- Nemani et al. (2023) on uncertainty quantification in machine learning

We have included a qualitative comparison of uncertainty proxies, noting that ambiguity adds incremental explanatory power beyond volatility and policy uncertainty in our setting. We avoid reporting separate comparison-table magnitudes to maintain consistency with the article. The literature additions also improve contextual grounding and demonstrate relevance across asset classes and regimes.

We believe these revisions have substantially strengthened the manuscript. The changes address all concerns raised by the reviewer while enhancing the paper's contribution to understanding uncertainty transmission in emerging markets. We thank the reviewer for their valuable input.

References

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